Python Cheatsheet

Essentials of Data Science in Python · Session 1 · Regression

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Everything you need to type, in one place.

Starting Python in VS Codium

Open the course folder once — **File > Open Folder**, then pick the repository. Everything below assumes you are inside it, and in **the VS Codium terminal** (**View > Terminal**, or `Ctrl` + `` ` ``) rather than the macOS Terminal or PowerShell.

Create the environment the first time only:

```
python3 -m venv .venv      # macOS
python3 -m venv .venv      # Linux
py -m venv .venv           # Windows (PowerShell)
```

Then activate it — **every time you open a new terminal**:

```
source .venv/bin/activate  # macOS
source .venv/bin/activate  # Linux
.venv\Scripts\activate     # Windows (PowerShell)
```

You will see `(.venv)` at the start of the prompt. If you do not, nothing below will find pandas. Install the packages once, with the environment active, and check the course data is intact:

```
pip install -r requirements.txt
python scripts/verify_puzzles.py    # 39 lines of [  ok  ], in about twenty seconds
```

Finally tell the editor which Python to use: `Ctrl`/`Cmd` + `Shift` + `P`, type **Python: Select Interpreter**, and choose the one inside `.venv`.

A new file, and running it

File > New File, then save it with a `.py` ending — `session01.py`, say. Write your code, save it, and in the terminal:

```
python session01.py     # macOS, Linux and Windows alike, inside .venv
```

Inside the environment the command is `python` on all three platforms. Outside it, macOS and Linux need `python3` and Windows needs `py`, which is one more reason to activate first. For trying one line at a time, type `python` on its own to get the `>>>` prompt, and `exit()` to leave.

The four things you always do

```
import numpy as np
import pandas as pd
import essentials as ess

# 1 - load a course dataset, or build a frame by hand
d = ess.load("coasters")                # generated locally; no network needed
ess.catalog()                              # the other five, and their puzzles
df = pd.DataFrame({"country": ["FR", "DE"], "gdp_pc_eur": [37_000, 46_000]})

# 2 - look at what is in it
d.head()           # the first five rows
d.shape            # (rows, columns)
```

```
d.dtypes            # what type each column holds
d.info()            # types, and how many values are missing
d.describe()        # count, mean, sd and quartiles, per numeric column

# 3 - read a file in
df = pd.read_csv("data.csv")
df = pd.read_excel("data.xlsx")             # needs openpyxl - see below
df = pd.read_parquet("data/coasters.parquet")

# 4 - write a CSV out
d.to_csv("out.csv", index=False)            # index=False, or you get a stray column
```

`import essentials as ess` works from the repository root. From a subfolder — a starter script in `01-regression/02-practice/starter/`, say — point Python back at the root first, which is what the two lines at the top of every starter do:

```
import sys
from pathlib import Path
sys.path.insert(0, str(Path(__file__).resolve().parents[3]))
import essentials as ess
```

Installing a package you do not have

Everything this course needs is already in `requirements.txt`. When you meet a `ModuleNotFoundError` anyway, or a line above says a reader needs something extra, install it **into the environment** — never outside it:

```
pip install openpyxl                # one package, by name
pip install -r requirements.txt     # or the whole course list again
```

Two rules, and both are about where it lands:

- Your prompt must show `(.venv)` **first**. `pip -V` prints the path it is about to install into; that path has to contain `.venv`.
- Type `pip`, not `pip3`, and never `sudo`. Inside an activated environment `pip` is already the environment's own.

Then check it arrived:

```
pip show openpyxl                   # name, version and location - or nothing at all
python -c "import openpyxl; print(openpyxl.__version__)"
```

What you have learned so far

Session 1, in the order you met it. Every line below has run on the course data.

Session 1 · Regression

```
# -- Load and shape -----------------------------------------------
import essentials as ess
d = ess.load("coasters")          # generated locally, no network
ess.catalog()                     # every dataset, and its puzzle
n_rows, n_cols = d.shape
d[["coaster", "drop_m", "speed_kmh"]].head()
g = d[d.propulsion == "gravity"].copy()
# .copy() before you modify a subset, or pandas warns and you guess
g["log_speed"] = np.log(g.speed_kmh)      # a power law becomes a line
c = np.exp(power.params["Intercept"])     # and back again
bands = pd.cut(g.drop_m, [0, 25, 45, 65, 85, 105, 200])
(g.assign(resid=lin.resid)
  .groupby(bands, observed=True)["resid"].agg(["mean", "count"]))
```

```
# the sign of the mean residual, band by band, is the whole diagnosis

# -- Describe ------------------------------------------------------
d.describe()                      # count, mean, sd, quartiles
d.propulsion.value_counts()       # how many of each

# -- Fit -----------------------------------------------------------
import statsmodels.formula.api as smf
lin = smf.ols("speed_kmh ~ drop_m", data=g).fit()
# read the ~ as "explained by"; the intercept is added for you
fit = model.fit()                 # .fit() does the arithmetic

# -- Read the fit --------------------------------------------------
lin.params["drop_m"], lin.bse["drop_m"]   # estimate, and its wobble
lin.tvalues["drop_m"], lin.pvalues["drop_m"]
lo, hi = lin.conf_int().loc["drop_m"]     # 95% by default
lin.rsquared, lin.rsquared_adj    # the second penalises extra columns
lin.resid, lin.fittedvalues       # what is left, and what was predicted
print(lin.summary().tables[1])    # the coefficient table on its own

# -- Diagnose ------------------------------------------------------
inf = lin.get_influence()         # the whole diagnostic bundle
h = inf.hat_matrix_diag           # leverage; it sums to p, always
(h > 2 * 2 / len(d)).sum()        # the 2p/n screen
cooks = inf.cooks_distance[0]     # leverage x residual; screen at 4/n

# -- Plot ----------------------------------------------------------
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(7.2, 3.2))
ax.scatter(g.drop_m, g.speed_kmh, s=20, alpha=0.8)
ax.set_xlabel("Drop (m)"); ax.set_ylabel("Top speed (km/h)")
# say the units, every time
ax.axhline(0, lw=1)               # the line you are judging against
ax.axhline(4 / len(d), ls="--", lw=1)     # the Cook's screen
ax.vlines(range(len(cooks)), 0, cooks, lw=1.4)   # one spike per row
plt.show()                        # last line of every figure
```